



Spotify Analytics & Machine Learning Project

A Comprehensive Data Science, BI, and ML Framework

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Chapter 1

Executive Summary

This project presents a comprehensive analytical framework for understanding Spotify's ecosystem by integrating user churn analysis, song popularity prediction, and personalized recommendation methodologies. Using Power BI, we conducted an in-depth examination of churn behavior across demographics, subscription categories, and user engagement patterns to identify the primary drivers of retention—without relying on machine learning models for this component.

The core predictive analytics were performed on Spotify song datasets, where multiple machine learning models were evaluated to forecast 2025 song popularity. Classification models, particularly CatBoost, achieved an accuracy of approximately 80%, significantly outperforming regression approaches that yielded an R^2 of roughly 51%. These results indicate that song popularity is better captured through categorical prediction rather than continuous regression.

Furthermore, a hybrid recommendation engine was developed by combining collaborative filtering with content-based KNN similarity. A tunable alpha parameter was introduced to dynamically adjust the balance between user-based interaction patterns and content-based audio similarity, resulting in highly accurate and flexible song recommendation capabilities.

Unsupervised learning techniques were also employed to explore musical structure and segmentation within the dataset. Hierarchical clustering revealed six meaningful song clusters, while K-Means produced eight distinct musical groupings, offering valuable insights for playlist curation and targeted marketing strategies.

Overall, this integrated analytical approach provides robust, actionable intelligence that supports improved user retention, enhanced engagement through personalized recommendations, and stronger forecasting of song success. The findings emphasize the strategic importance of combining business intelligence, predictive modeling, and recommender systems to drive data-informed decision-making within the music streaming industry.

Chapter 2

Introduction

Context

With millions of songs available on streaming platforms, users often struggle to discover relevant content, while artists and producers aim to understand what drives a song's popularity.

Problem Statement

How can we predict whether a song will become popular based on its characteristics?
How can we recommend songs similar to a given one to improve listener engagement?

Objectives

Analyze, present insights and findings, and visualize Spotify data using Power BI and python.

- Build classification models to predict song popularity.

- Develop a hybrid recommender system that uses both KNN similarity and collaborative filtering.

Expected Outcomes

Insights into the drivers of music popularity.

- A working recommendation pipeline producing relevant playlist (k songs) suggestions.

Chapter 3

Sources

Source 1: Spotify 30,000 Songs Dataset *Link: <https://example.com>* This dataset offers a large set of structured audio features (danceability, energy, valence, etc.). It was chosen because its size supports reliable statistical analysis, and its standardized format makes it ideal for machine learning and recommendation modeling.

Source 2: Churn Dataset (Synthetic) *Link: <https://example.com>* Synthetic churn data was used to model user retention without exposing sensitive personal information. It allows controlled experimentation with classification algorithms and helps evaluate whether listening behavior correlates with churn risk.

Source 3: Spotify API (Top Arabic Hits + Artists: Eminem, NF, Taylor Swift, The Weeknd – Popularity 2025) *Link: <https://example.com>* The API provides real-time and dynamic music data, enriching the static datasets with fresh trends. It was selected to show the integration of live external sources and to make recommendations more realistic and time-aware.

Source 4: GDP and Happiness Index (Web Scraping) *Link: <https://example.com>* These socioeconomic indicators were collected to explore potential links between external context and listening habits. Web scraping ensures up-to-date values and demonstrates automation of data extraction.

Chapter 4

Tools

Python / Jupyter Notebook Used for the entire analytical pipeline: data cleaning, exploratory analysis, visualization, and machine learning. Python’s ecosystem (Pandas, Matplotlib, Scikit-learn, Requests) supports structured workflows combining local datasets, APIs, and scraped data.

Power BI Chosen for building interactive and decision-friendly dashboards. Its DAX calculations and visualization capabilities translate complex analytical findings into clear performance indicators.

L^AT_EX Used for producing a structured, professional-quality report. It ensures clean formatting, consistent structure, and high-quality presentation of technical and analytical content.

Chapter 5

Data Collection / Extraction & Understanding

Churn Dataset (Power BI Part)

Rows: Each row represents a unique Spotify user.

Columns (Features):

user_id → Unique identifier for each user

gender → User gender (Male/Female/Other)

age → User age

country → User location

subscription_type → Type of Spotify subscription (Free, Premium, Family, Student)

listening_time → Minutes spent listening per day

songs_played_per_day → Number of songs played daily

skip_rate → Percentage of songs skipped

device_type → Device used (Mobile, Desktop, Web)

ads_listened_per_week → Number of ads heard per week

offline_listening → Offline mode usage

is_churned → Target variable (0 = Active, 1 = Churned)

Dataset Type: Mixed (numeric + categorical)

30000 Songs Dataset : Spotify Songs Features

track_id: A unique identifier assigned to each song.

track_name: The title or name of the song.

track_artist: The name of the artist who performed the song.

track_popularity: A score ranging from 0 to 100 indicating the popularity of the song.

track_album_id: A unique identifier associated with the album.

track_album_name: The name of the album in which the song was released.

track_album_release_date: The release date.

playlist_name: Playlist name.

playlist_id: Unique playlist identifier.

playlist_genre: Genre category.

playlist_subgenre: More specific subgenre.

danceability: 0–1 measure of dance suitability.

energy: 0–1 measure of intensity.
key: Musical key (0–11 or -1).
loudness: dB level.
mode: 1 = major, 0 = minor.
speechiness: 0–1 presence of spoken words.
acousticness: 0–1 acoustic measure.
instrumentalness: 0–1 probability of no vocals.
liveness: Probability of being recorded live.
valence: Positive emotional tone.
tempo: BPM.
duration_ms: Length in ms.

Happiness Index and GDP Dataset (Web Scraping)

To enrich the dataset with socioeconomic indicators, we programmatically extracted GDP and happiness index data using requests and `pandas.read_html()`. Irrelevant columns were removed, country names cleaned, missing values handled, and CSV files generated.

Spotify API

1. Enhancement to Main Songs Dataset

A new feature called `popularity_2025` was added to the original dataset (which had only `popularity_2023`).

2. Artist-Specific Datasets Collected

Arabic Hits Dataset: track name, ID, popularity, artist info.

Eminem Dataset: track name, album, popularity, duration, release year.

NF Dataset: track name, album, popularity, duration, explicit flag.

Taylor Swift and The Weeknd: extracted but not used.

3. Purpose and Platform Usage Summary

Arabic Hits, Eminem, and NF used for artist-specific trends.

Eminem used in Python and Power BI.

NF used only in Power BI.

`popularity_2025` integrated for prediction.

Taylor Swift and The Weeknd excluded.

4. API Acquisition Methodology

Automated pipeline using Spotify Web API, OAuth 2.0, token generation, modular functions, batch playlist extraction, global chart extraction, and popularity updates.

Authentication: `CLIENT_ID` and `CLIENT_SECRET` stored in `.env`, `get_access_token()` handles token.

Artist extraction: `get_artist_id()`, `get_artist_albums()`, `get_album_songs()`, `get_track_details()`, `create_albums_tracks_data()`.

Global charts: playlist IDs, `get_playlist()`.

Popularity 2025: `get_popularity_scores()`.

Why effective: scalable, secure, consistent, real-time capable, JSON structured.

Chapter 6

Data Cleaning & Preparation

4.1 Data Cleaning

Removed rows with null names/dates (1200). Dropped unnecessary columns (album_id, song_id). Removed duplicates and handled missing values. Corrected date column types.

4.2 Feature Engineering

Added popularity_2025. Created new features: season, period, number_of_songs_per_artist, release_age, year. Standardized album release date column.

4.3 Dataset Structuring

Two datasets generated: grouped-by-artist and full song-level dataset.

4.4 Data Transformation

Normalization of danceability, energy, loudness, tempo. Categorical encoding of artist and genre.

Popularity label conversion:

0–30: hardly recognized 30–60: famous 60–100: hit

Creation of user–song interaction matrix.

Train/test split (80/20).

Chapter 7

Exploratory Data Analysis (EDA)

1) Churn Dataset – Exploratory Data Analysis

The exploratory analysis conducted on the churn dataset included:

- Correlation heatmap to examine relationships between user attributes and churn.
- Distribution plots for all numerical variables.
- Normality curves to assess the distribution of variables.
- Box plots for each feature to detect outliers and variance.
- Scatter plots for pairwise feature relationships.
- Pair plot for multivariate distribution and interaction exploration.

Main Observation:

The dataset proved to be **not meaningful**, showing **no strong patterns or interesting relationships** with the target variable (is_churned). This lack of structure hindered deeper analysis and prevented reliable predictive modeling.

2) Songs Dataset – Exploratory Data Analysis

A comprehensive EDA was performed on the 30,000-song dataset, including:

- Correlation heatmap to evaluate relationships between audio and metadata features.
- Distribution plots for all numerical variables.
- Normality curves to detect skewness and distribution shapes.
- Box plots for each column to identify variance and outliers.
- Scatter plots and pair plot to visualize multivariate feature interactions.
- Histograms for all numerical audio features.
- Regression plots showing key correlations:

- Energy and acousticness — **negative correlation**
- Loudness and energy — **positive correlation**
- Valence and instrumentalness — **positive correlation**
- Genre and subgenre distribution analysis.
- Identification of:
 - Top 10 most influential features
 - Top albums
 - Top playlists
 - Top tracks
- Time-based trend analyses:
 - Number of tracks per period
 - Seasonal popularity trends
 - Songs popularity area plot
 - Average popularity per period per song
 - Average popularity of song genres per year (line plot)

Main Focus of This Analysis:

The goal was to understand how **audio characteristics**, **genre classifications**, and **temporal variables** influence popularity, and to extract patterns useful for the recommendation system and popularity prediction models.

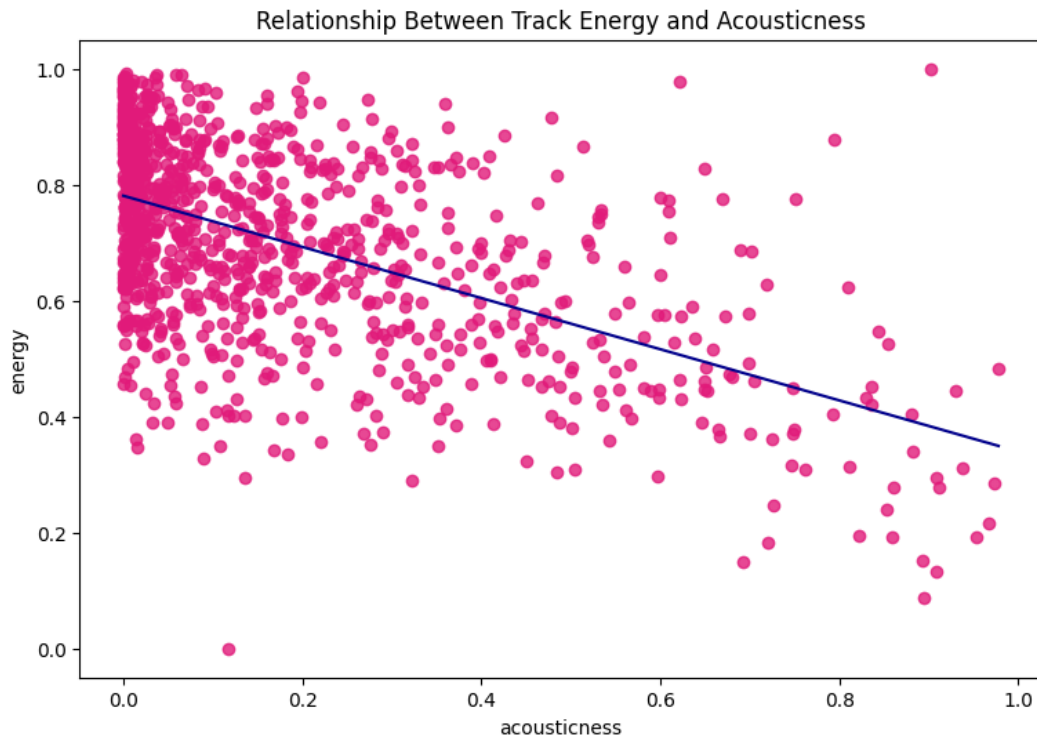


Figure 7.1: Relationship between energy and acousticness

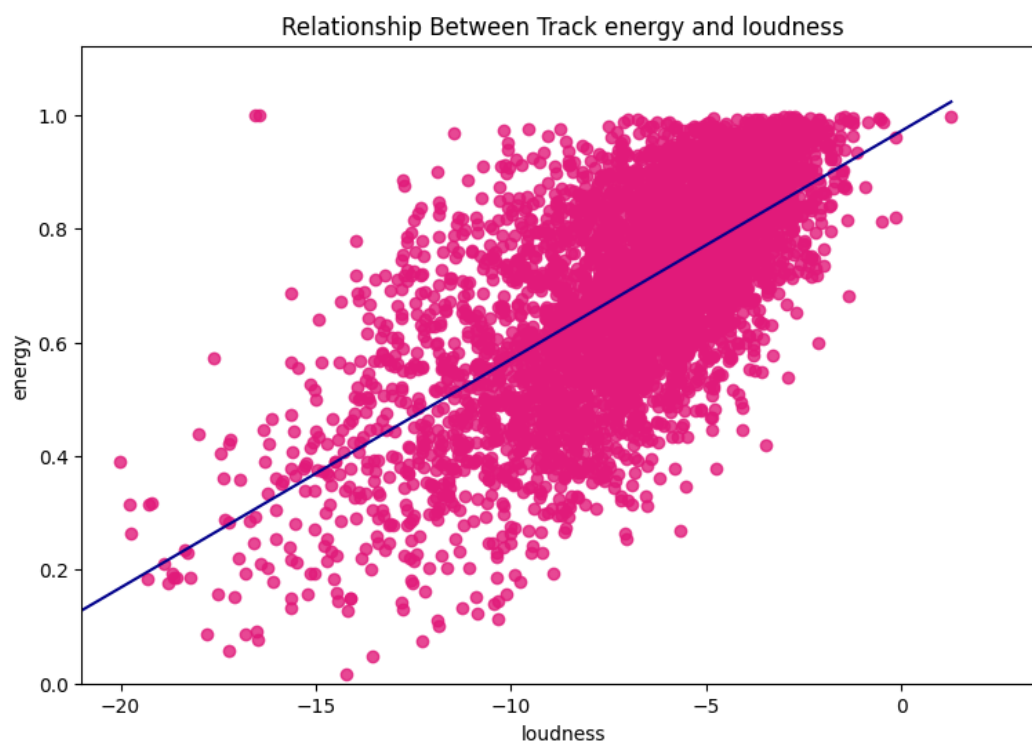


Figure 7.2: Relationship between energy and loudness

Audio Features Relationship with Song Popularity in 2025

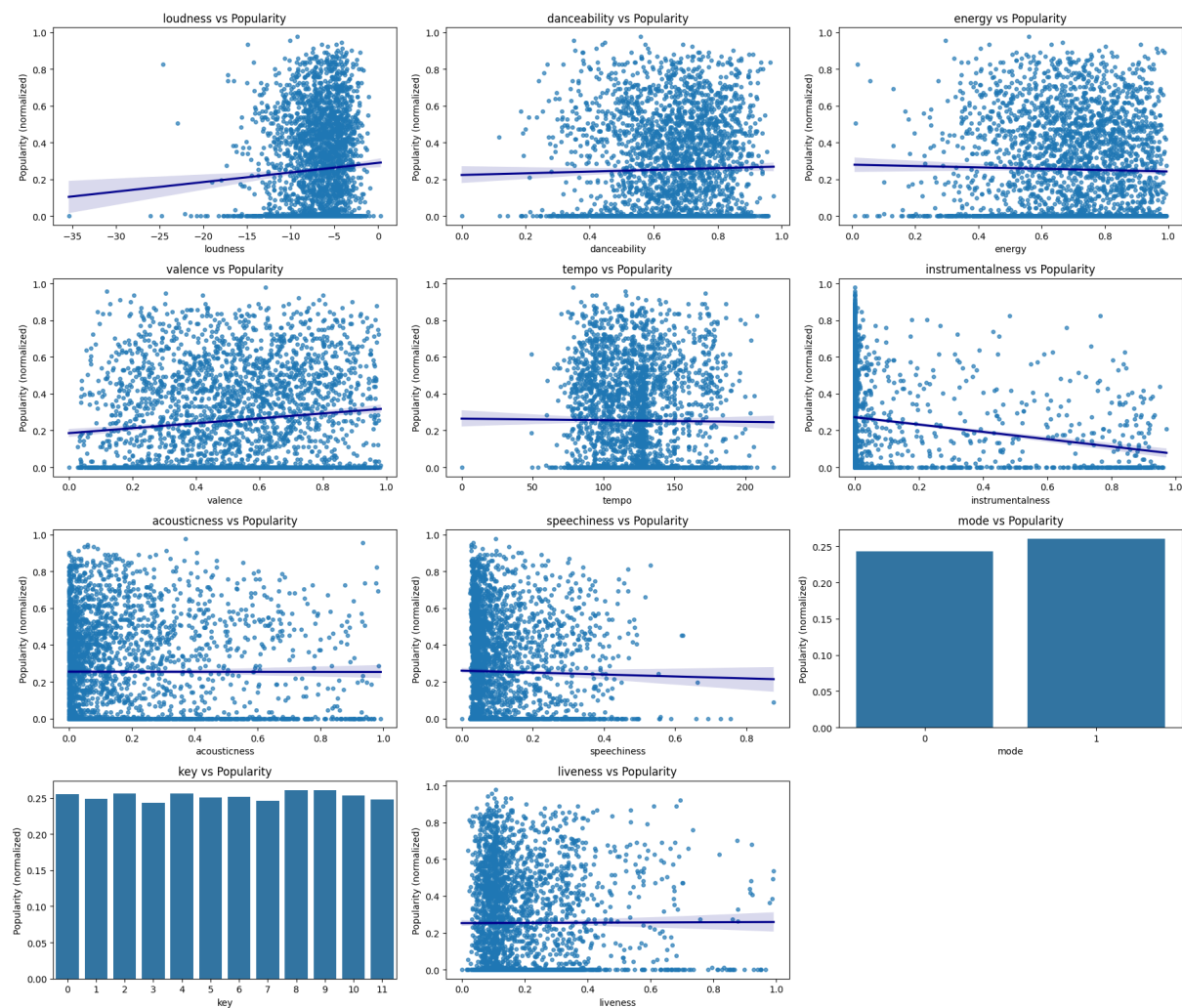


Figure 7.3: Audio features vs popularity (multi-plot)

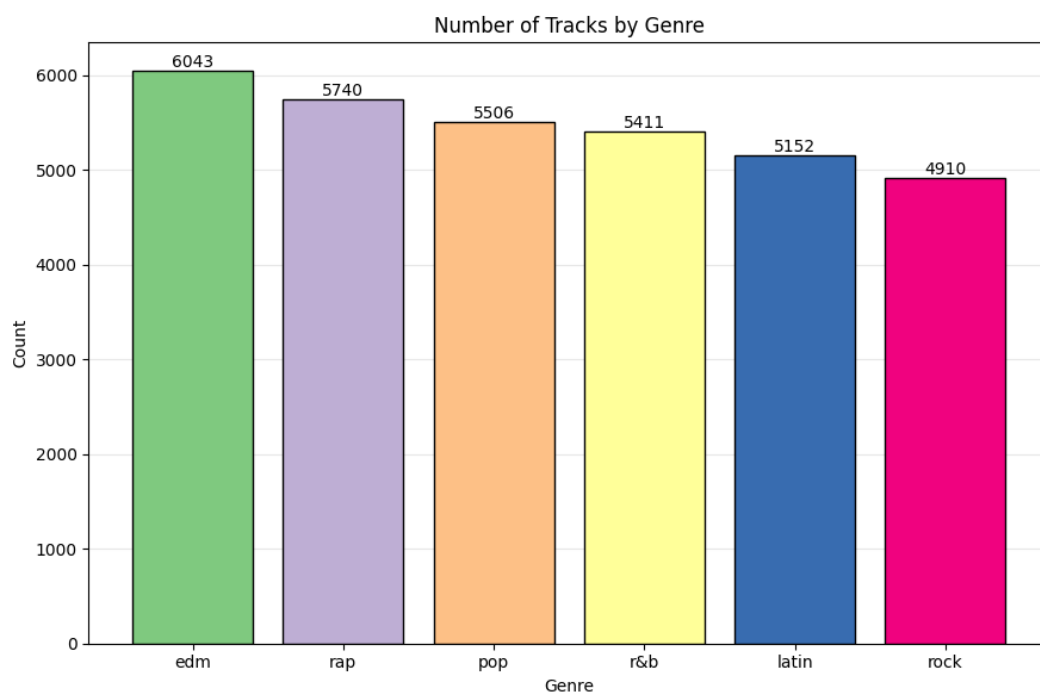


Figure 7.4: Number of tracks by genre

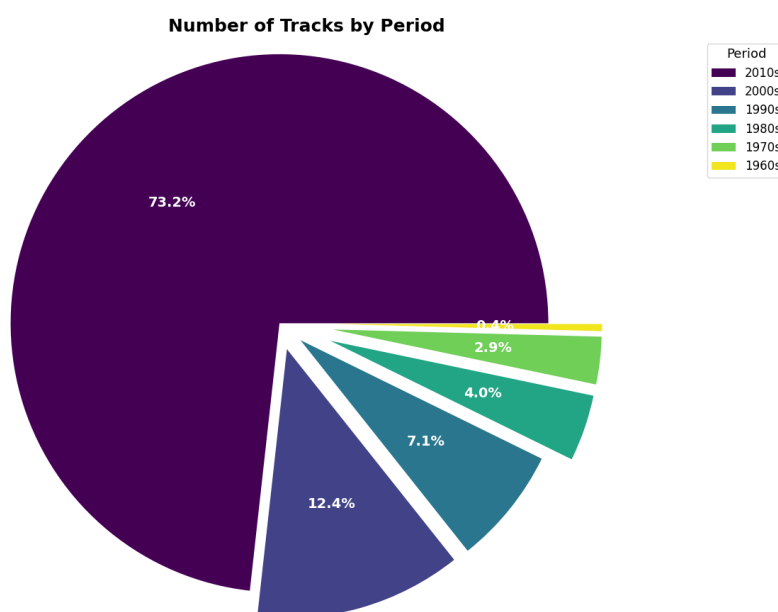


Figure 7.5: Proportion of tracks by period

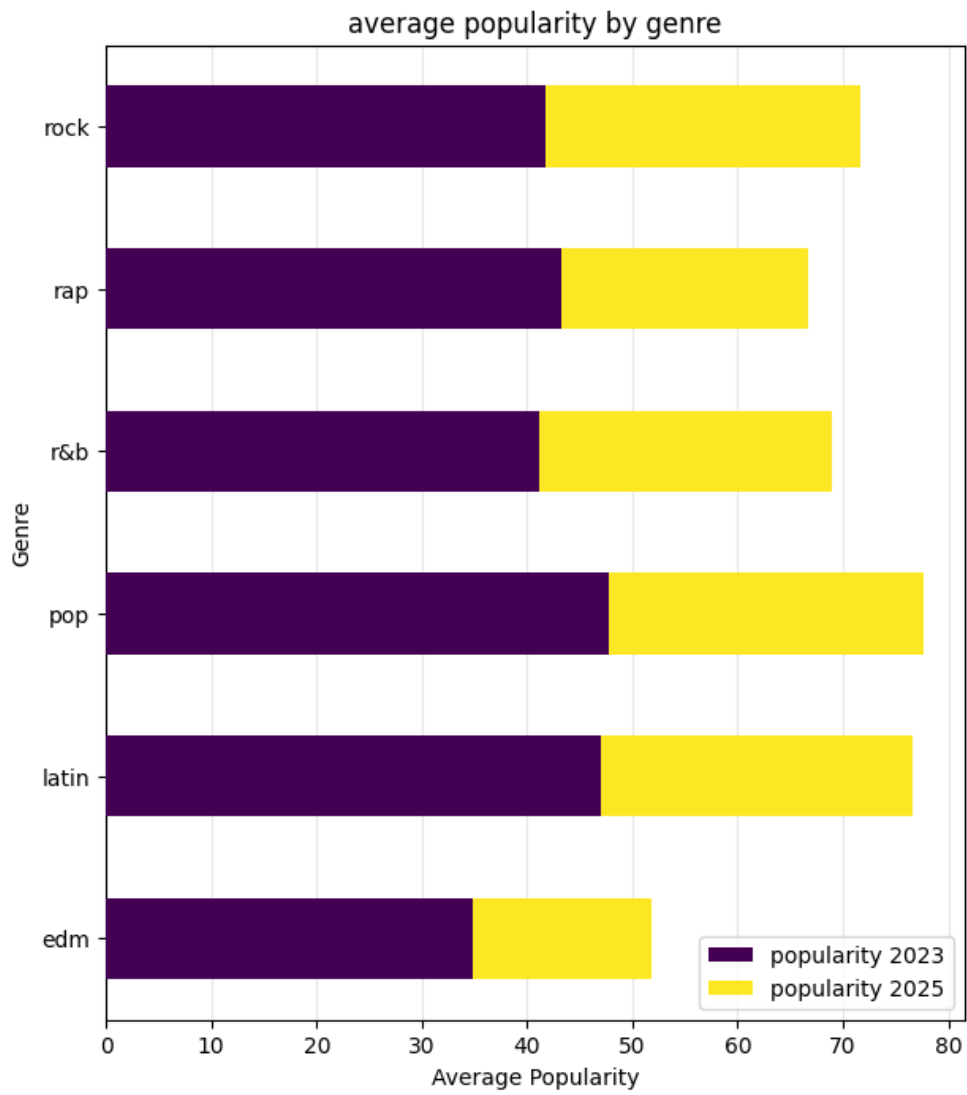


Figure 7.6: Average popularity by genre (2023 vs 2025)

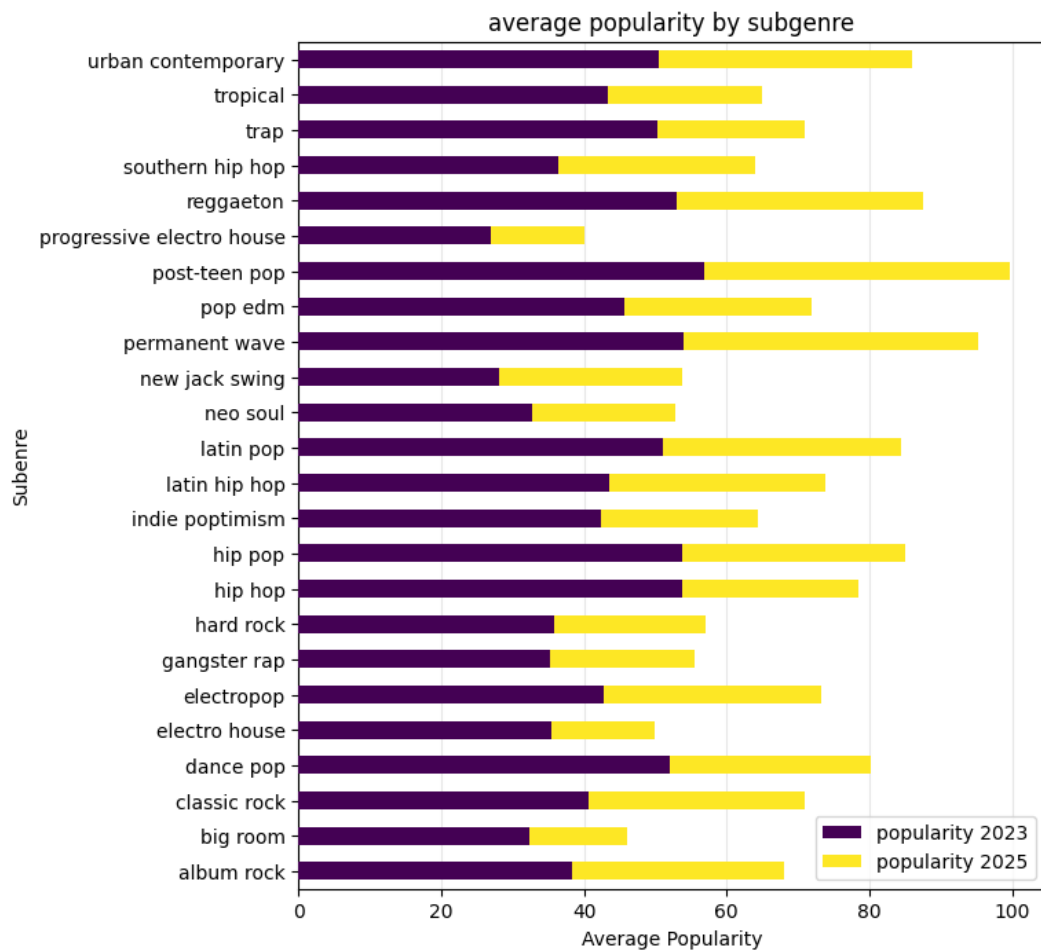


Figure 7.7: Average popularity by subgenre (2023 vs 2025)

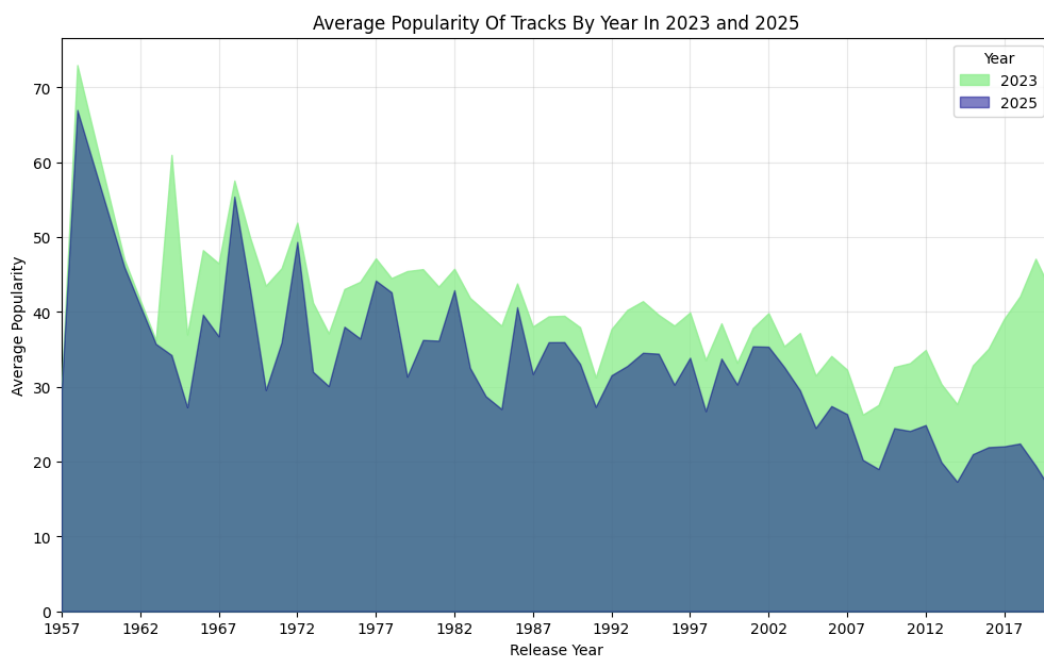


Figure 7.8: Average popularity of tracks by year (2023 vs 2025)

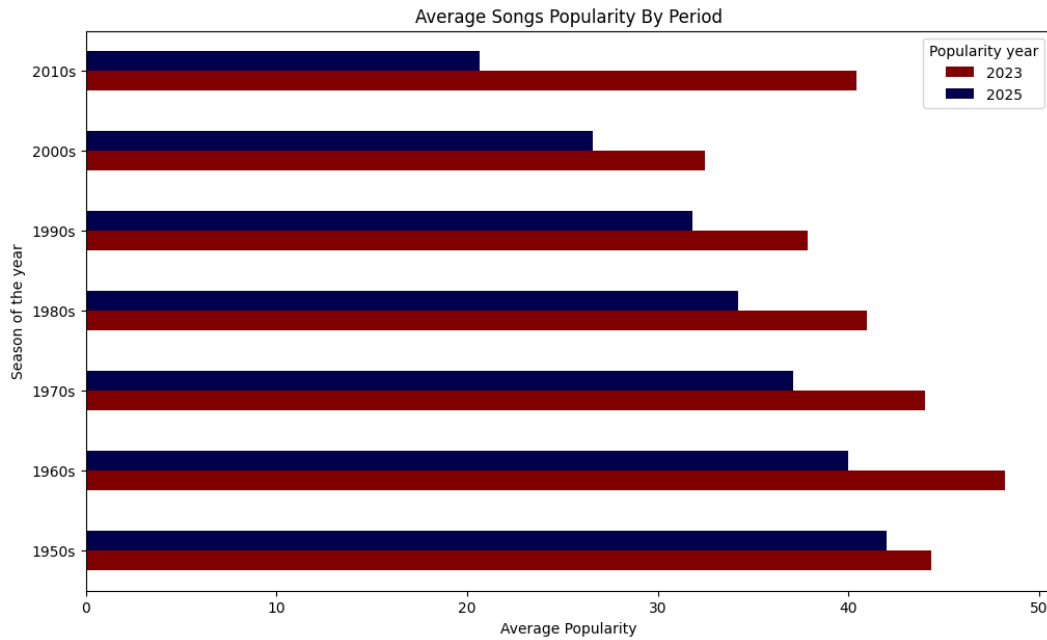


Figure 7.9: Average songs popularity by period (2023 vs 2025)

3) Artist-Level Analysis

3.1 Eminem Dataset

The analysis for Eminem included:

- Top 20 most popular tracks in 2025.
- Bubble plot showing album track count vs. average popularity.
- Average popularity of Eminem songs by release year.
- Word cloud of song titles scaled by popularity.

Purpose: To understand artist-specific performance patterns, popularity evolution, and relationships between album structure and listener engagement.

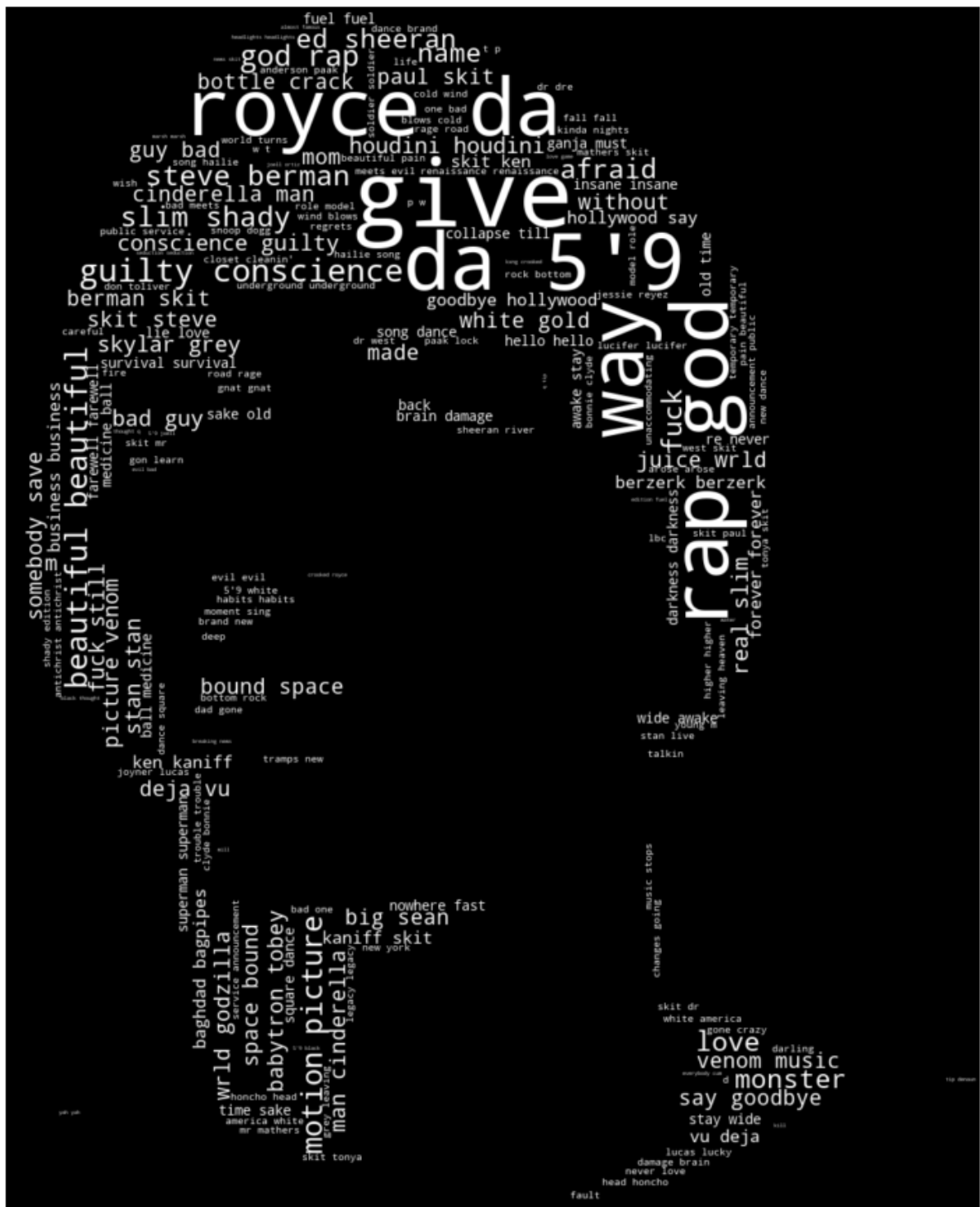


Figure 7.10: Wordcloud of Eminem song titles Weighed by popularity in 2025

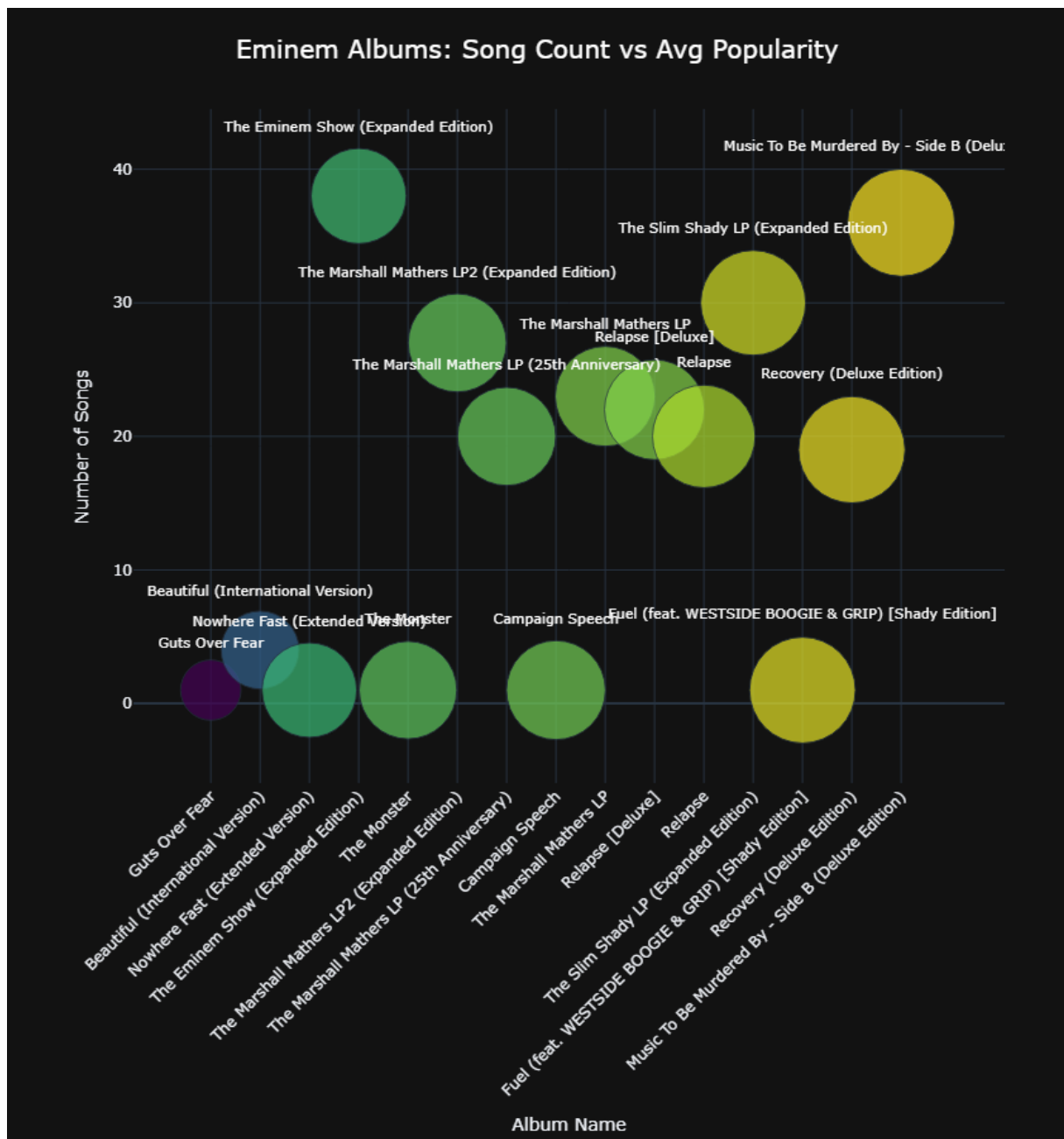


Figure 7.11: Bubble plot of Eminem Albums vs Songs Count vs Average Song Popularity

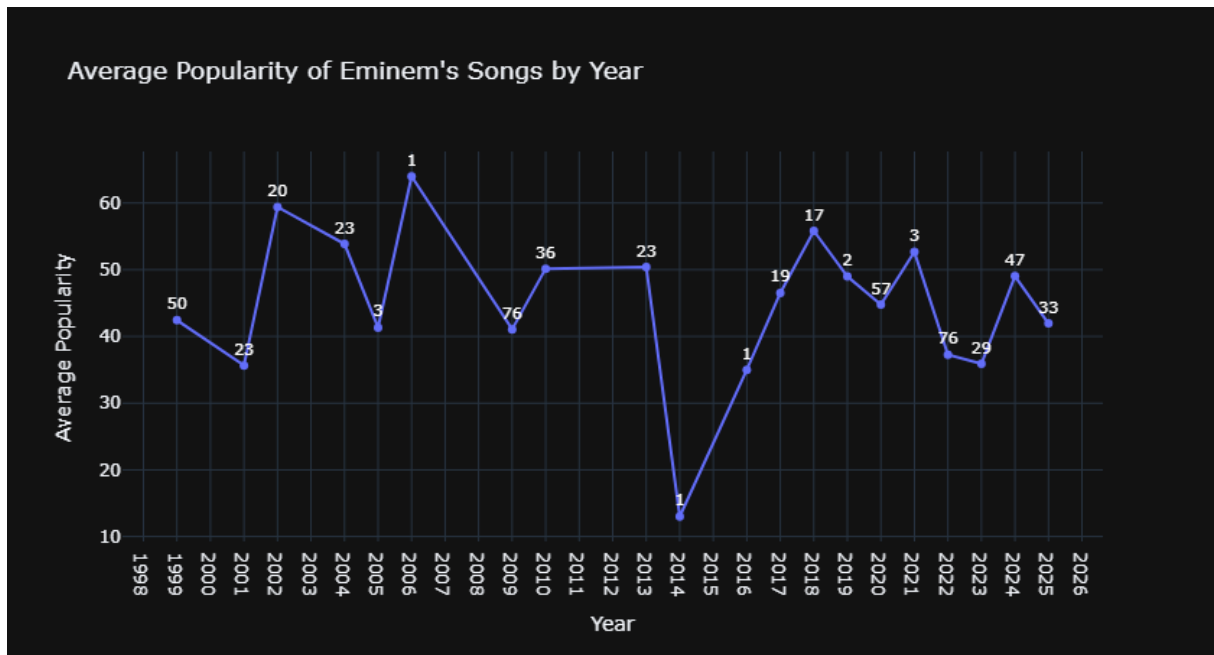


Figure 7.12: Average Popularity of Eminem's Song by Year

3.2 Arabic Hits Dataset

The Arabic Hits analysis included:

- Top 20 most popular tracks in Arabic countries on Spotify.

Purpose: To identify dominant artists, regional listening trends, and structural patterns within Arabic music markets.

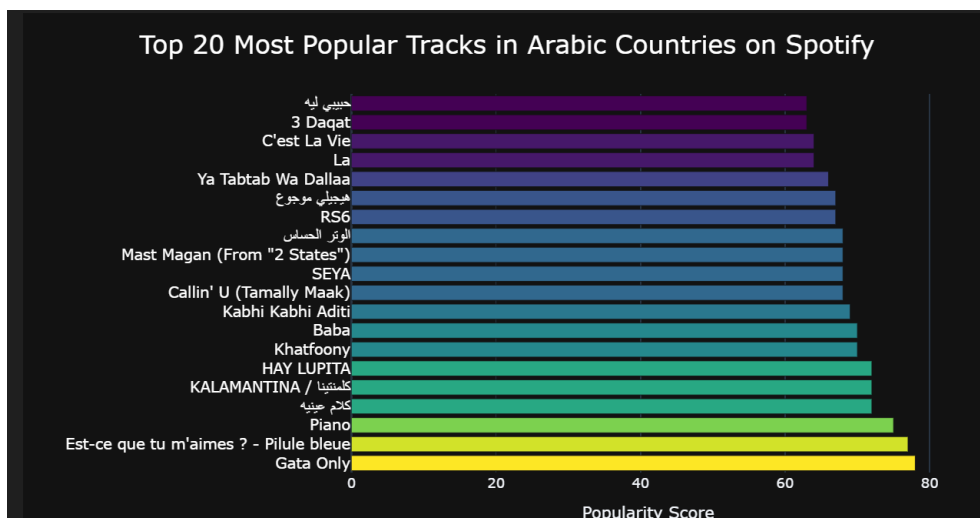


Figure 7.13: 2025 Top 20 arabic hits by popularity score

Chapter 8

Machine Learning

8.1 Popularity Prediction

ML Part 1: Popularity Prediction

We initially attempted regression models (Random Forest, CatBoost and MLP Neural Network) on continuous `popularity_2025`, but achieved modest performance ($R^2 \approx 0.47\text{--}0.51$). This suggested that exact scores are difficult to model.

We then switched to a classification approach by converting popularity into three categories (hardly recognisable, famous, hit). Models used include:

- Random Forest
- CatBoost
- MLP Neural Network
- Logistic Regression

CatBoost and Logistic Regression followed by Artificial Neural Network(ANN/MLP) delivered the best results with macro-averaged AUC up to around 0.88 and accuracy between 0.73 and 0.88 depending on thresholds. Random Forest was slightly less accurate but useful for feature importance.

The analysis confirms that:

- popularity is better treated as a classification problem,
- artist-related variables and recency (`release_age`) are crucial drivers,
- energy, valence and danceability also play significant roles.

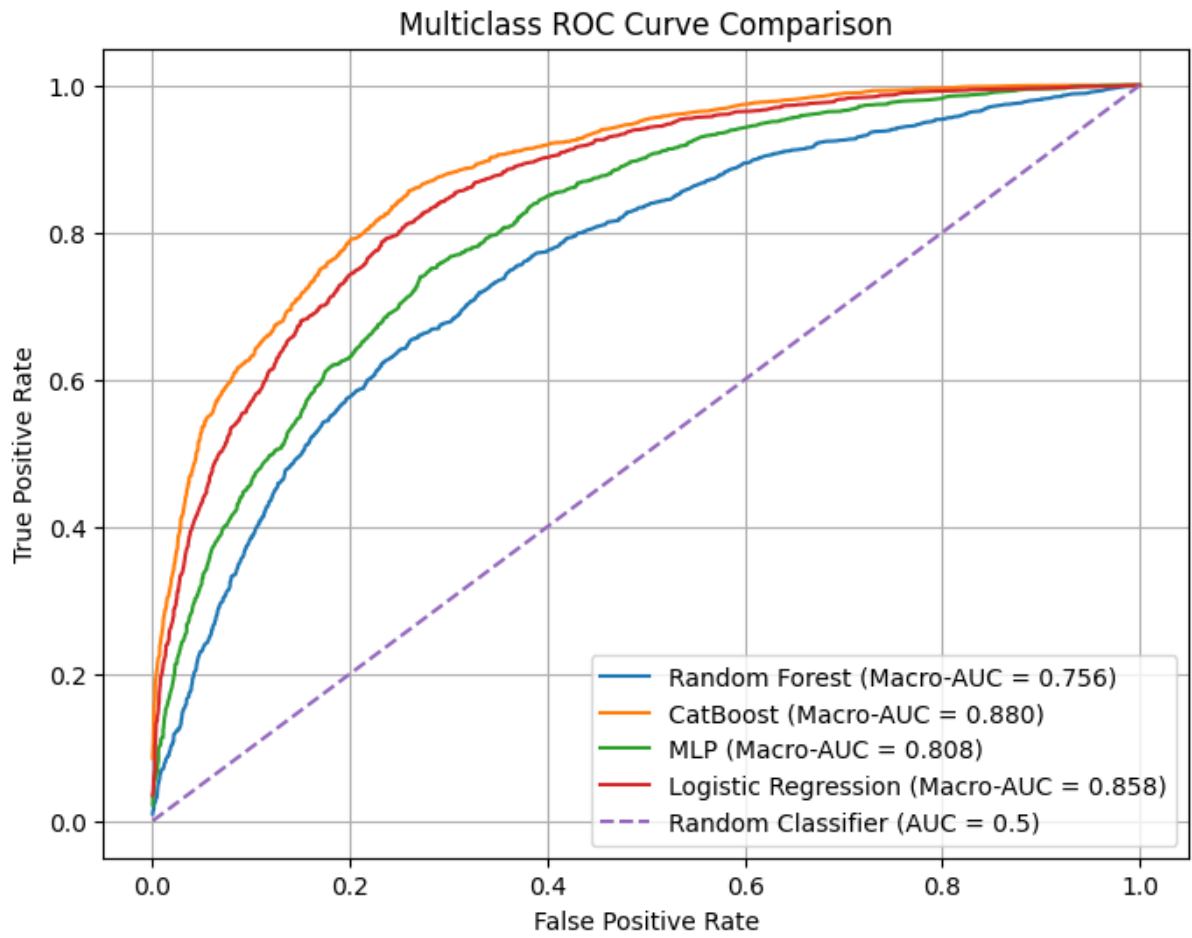


Figure 8.1: ROC-AUC analysis Curves

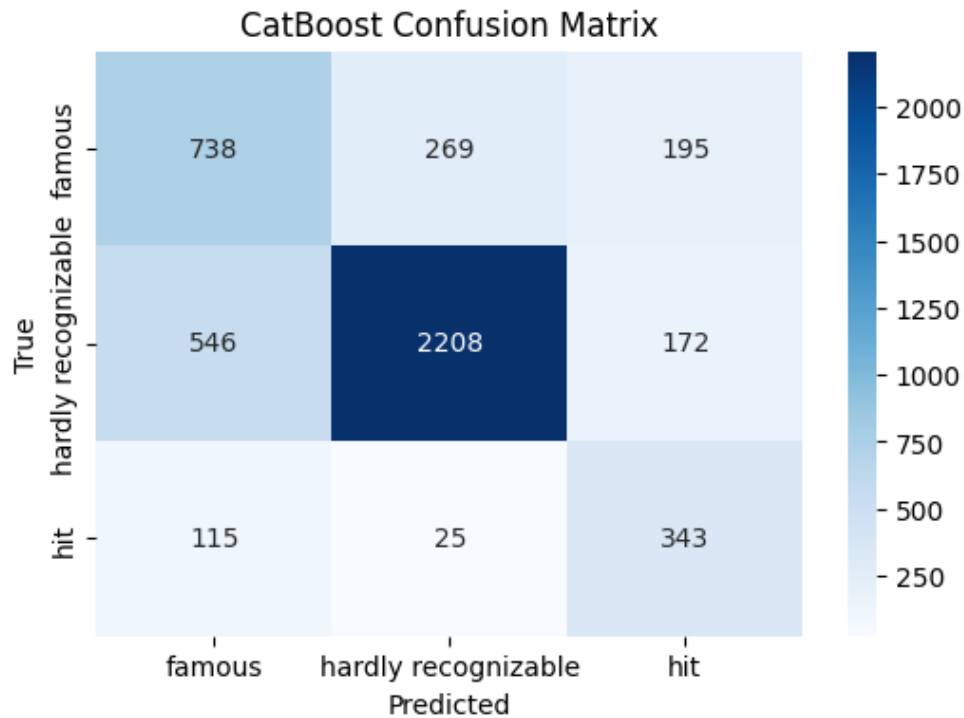


Figure 8.2: Confusion Matrix of the CatBoost Classification model

6.1 Song Popularity Prediction Models (Detailed Summary)

Four models were trained to predict popularity category (hit vs famous vs hardly recognizable).

Model	Accuracy	Notes
CatBoost	88%	Captures nonlinear interactions efficiently.
Logistic Regression	85%	Strong linear baseline.
MLP Neural Network	80%	Limited by tuning complexity.
Random Forest	75%	Most interpretable; weaker precision.

Feature Importance (Random Forest)

8.2 Recommendation System

ML Part 2: Recommendation System

1. Data Preprocessing

The first step in building our recommendation system involved preparing the dataset for machine learning models:

Rank	Feature	Importance	Interpretation
1	artist_avg_pop	0.213	Strong fanbase drives immediate popularity.
2	release_age	0.065	Recency bias: newer songs perform better.
3	artist_song_count	0.064	Larger catalogs → higher exposure.

Feature Engineering:

Categorical Features: Playlist genres and subgenres were originally stored as semicolon-separated strings. We split them into lists (`genre_list`, `subgenre_list`) and applied `MultiLabelBinarizer` to encode multiple categories into a binary format. Other categorical variables like `period` and `season` were encoded using one-hot encoding.

Numerical Features: Audio features such as `danceability`, `energy`, `loudness`, `tempo`, `valence`, and track popularity (`popularity_2025`) were normalized using `MinMaxScaler` to scale them between 0 and 1.

Feature Matrix: All numerical and categorical features were combined into a single feature matrix `X` for downstream analysis. Feature names were stored for interpretability.

2. Dimensionality Reduction with PCA

High-dimensional feature spaces can be noisy and computationally expensive. To address this, we applied Principal Component Analysis (PCA):

PCA Projection:

PCA was fitted on the combined feature matrix.

The first two principal components were visualized in a scatter plot to observe clustering tendencies in the data.

A scree plot was created to visualize the variance explained by each component.

The number of components explaining 90% of the variance (`n_components_90`) was retained for further analysis, reducing dimensionality while preserving most information.

Feature Importance in PCA:

By examining the absolute values of PCA coefficients, we identified which features contributed most to the principal components, providing interpretability of the feature space.

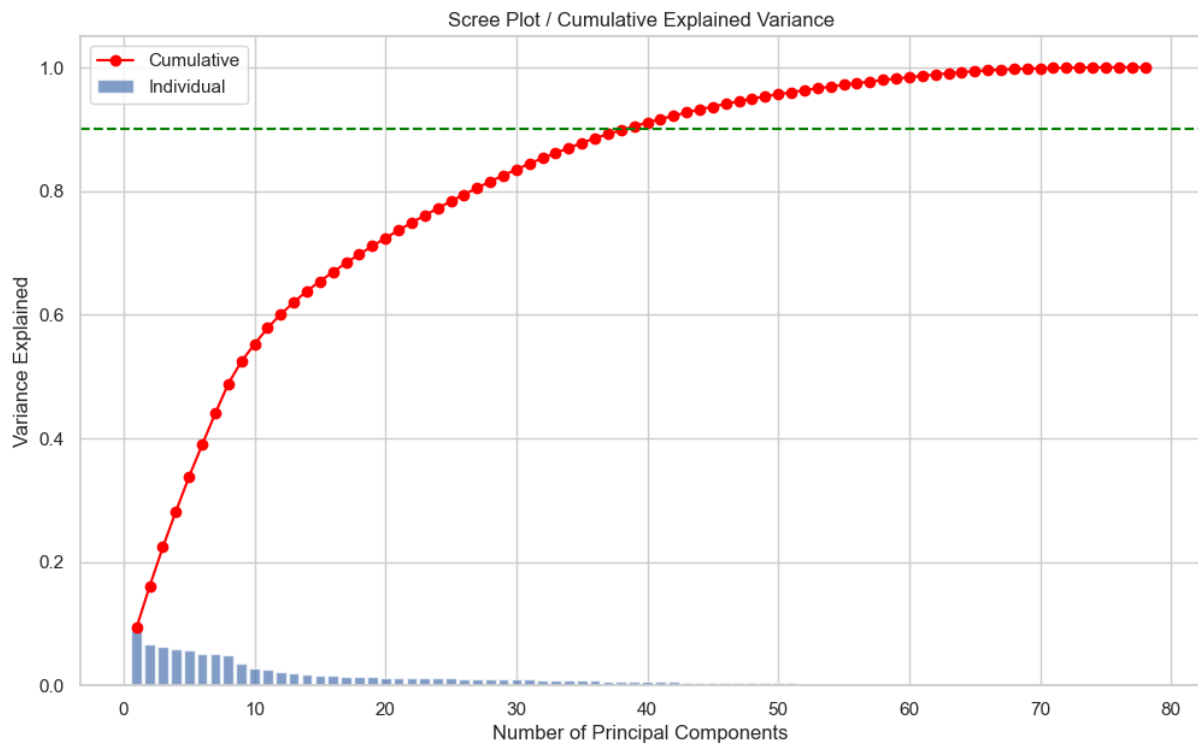


Figure 8.3: Screeplot of cumulative explained variance (PCA)

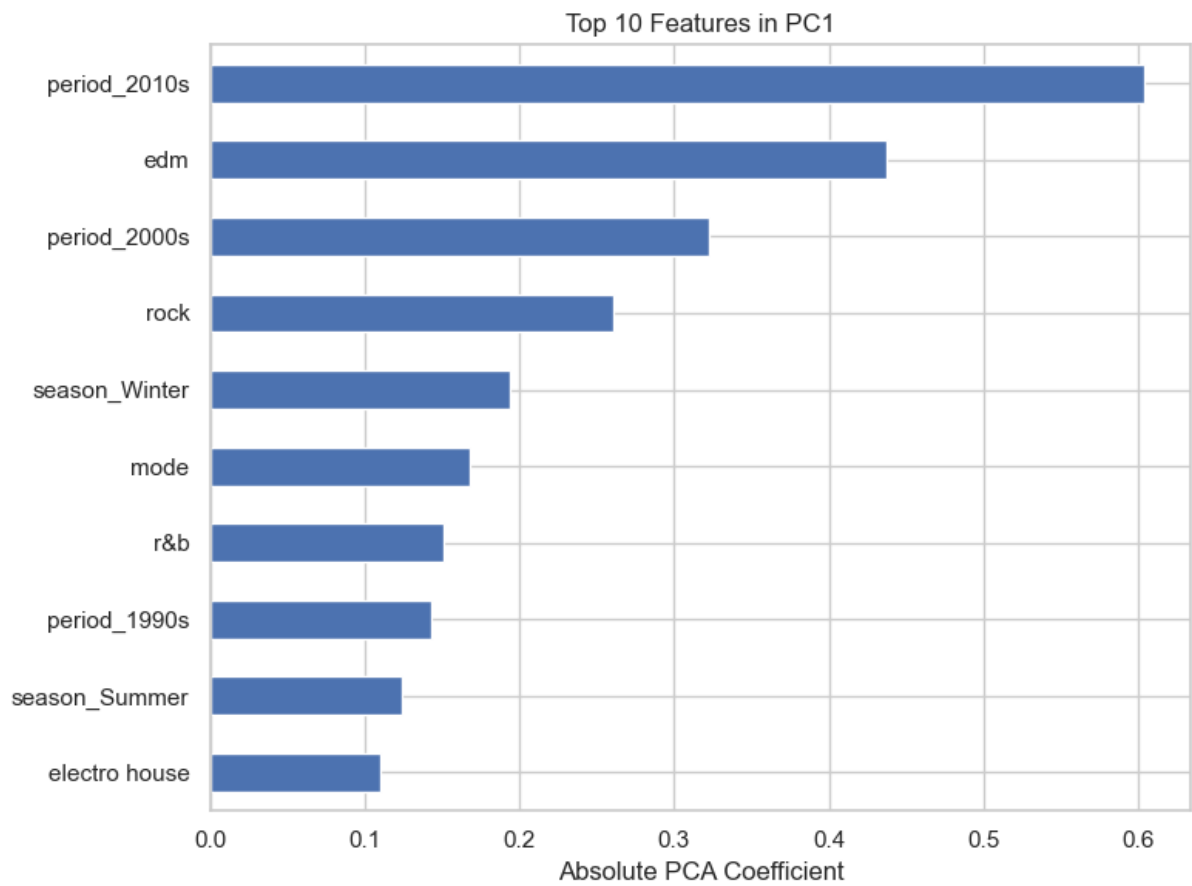


Figure 8.4: Top 10 important (high variance) features in the first principal component

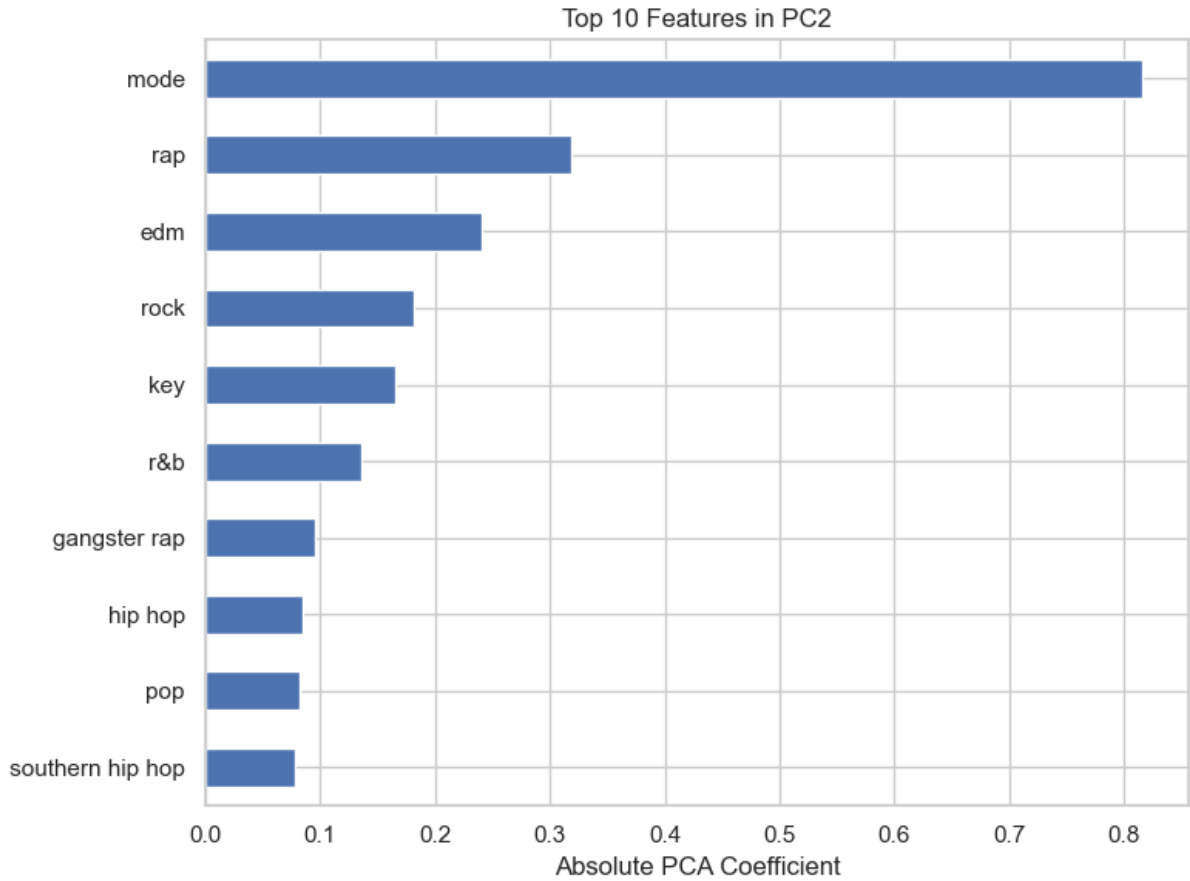


Figure 8.5: Top 10 important (high variance) features in the second principal component

3. Content-Based Recommendation using K-Nearest Neighbors

With the reduced PCA features, a content-based recommendation system was built:

KNN Model:

Metric: Cosine similarity was used to measure similarity between tracks in the PCA-reduced space.

Hyperparameter Selection: The optimal number of neighbors k was determined by testing average distances to nearest neighbors. The value minimizing the average distance was selected.

The trained KNN model allows finding tracks similar to a given track based on audio features and categorical metadata.

Example Recommendation:

For a given track, the model identifies its k most similar tracks.

Similarity scores are computed as:

$$1 - \text{cosine distance}$$

4. Collaborative Filtering via Playlist Co-Occurrence

In addition to content-based recommendations, we incorporated collaborative filtering to leverage user/playlist behavior:

Interaction Matrix:

Rows represent playlists, columns represent tracks, and values correspond to track popularity in the playlist (or binary presence).

Track Similarity:

Cosine similarity was computed between tracks based on co-occurrence across playlists.

This captures relationships between tracks that frequently appear together in user playlists.

5. Hybrid Recommendation

To combine the strengths of both content-based and collaborative filtering approaches:

Hybrid Score:

Weighted combination of content similarity (from KNN on PCA features) and collaborative filtering similarity (from playlist co-occurrence).

Parameter alpha controls the relative weight of content vs. collaborative similarity.

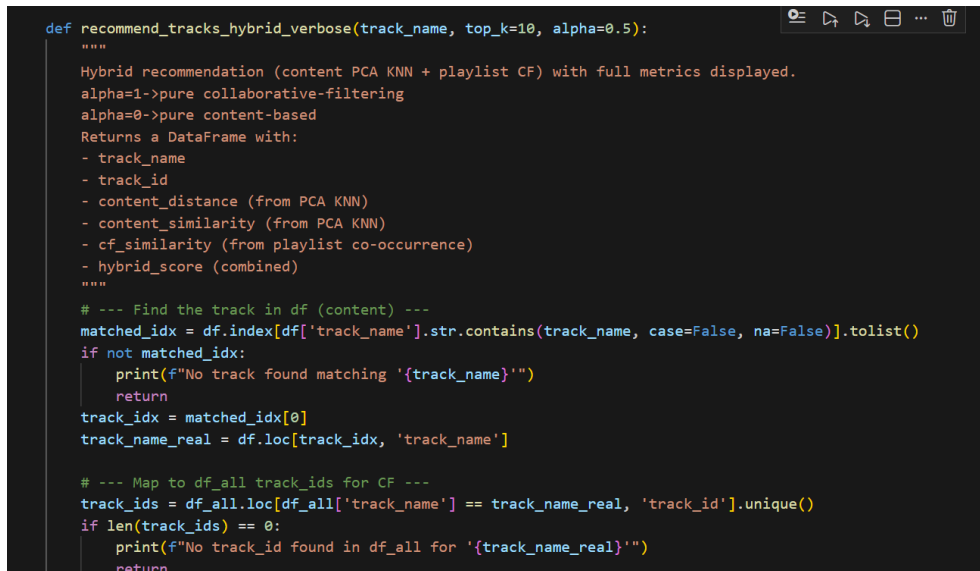
Recommendation Output:

For a given track, the top k recommended tracks are returned with:

Content similarity

CF similarity

Hybrid score



```
def recommend_tracks_hybrid_verbose(track_name, top_k=10, alpha=0.5):
    """
    Hybrid recommendation (content PCA KNN + playlist CF) with full metrics displayed.
    alpha=1->pure collaborative-filtering
    alpha=0->pure content-based
    Returns a DataFrame with:
    - track_name
    - track_id
    - content_distance (from PCA KNN)
    - content_similarity (from PCA KNN)
    - cf_similarity (from playlist co-occurrence)
    - hybrid_score (combined)
    """
    # --- Find the track in df (content) ---
    matched_idx = df.index[df['track_name'].str.contains(track_name, case=False, na=False)].tolist()
    if not matched_idx:
        print(f"No track found matching '{track_name}'")
        return
    track_idx = matched_idx[0]
    track_name_real = df.loc[track_idx, 'track_name']

    # --- Map to df_all track_ids for CF ---
    track_ids = df_all.loc[df_all['track_name'] == track_name_real, 'track_id'].unique()
    if len(track_ids) == 0:
        print(f"No track_id found in df_all for '{track_name_real}'")
        return
```

Figure 8.6: Partial Code Snippet for recommend tracks hybrid verbose function

```
recommend_tracks_hybrid_verbose("the search",top_k=10,alpha=1)
```

Top 10 hybrid recommendations for 'the search is over':

track_id	track_name	content_similarity	cf_similarity	hybrid_score
10b7RJ4vWHQExpTikfsf6A	the search is over	1.000000	1.0	2.000000
6BtlcfCWVBUEZwFaNJ0QMO	i can't hold back	0.898626	1.0	1.898626
6LDRtOMCSx81hKXAk1LuHd	at this moment	0.865703	1.0	1.865703
3LVRM23d9ltvRoquuZy1Yb	love touch - 2008 remaster	0.284574	1.0	1.284574
4ybDW0xfRBCPwsjwFOUzZW	no more lonely nights	0.016744	1.0	1.016744
1cOj24fUMnBSXhtp7waHVW	that's what friends are for	NaN	1.0	1.000000
7LoGfKBAAOl0nxhodJ1240	word up	NaN	1.0	1.000000
6yJxCltgtmGxlgKOS6ndnu	greatest love of all	NaN	1.0	1.000000
1IJZsMoWEbrgaY6CxPMf8	would i lie to you? - remastered version	NaN	1.0	1.000000
3FdHgoJbH3DXNtGLh56pFu	conga	NaN	1.0	1.000000
4H3vuLX59XPqdtTplesGyS	california girls	NaN	1.0	1.000000

Figure 8.7: Usage Example

6. Clustering for Song Analysis

Clustering was added to understand the underlying structure of the music dataset and enhance the recommendation system:

Clustering Algorithms:

K-Means: Optimal k selected using silhouette score. Captures general grouping of tracks based on audio features.

Agglomerative (Hierarchical) Clustering: Also optimized by silhouette score. Provides a hierarchical view of clusters, which can capture more nuanced groupings.

DBSCAN (optional): Density-based clustering to identify outlier tracks that do not fit well into standard clusters.

Integration with PCA:

Clustering was performed both on the high-dimensional PCA-reduced space (X_pca_90) and on 39D (78D→39D) PCA projections for visualization.

Clusters were visualized with scatter plots, showing the distribution of tracks in the reduced feature space.

Purpose of Clustering in the Project:

Identifies natural groupings of tracks based on audio and metadata features.

Enhances interpretability of recommendation results.

Can be used as an additional feature in hybrid recommendation (e.g., suggesting tracks from the same cluster).

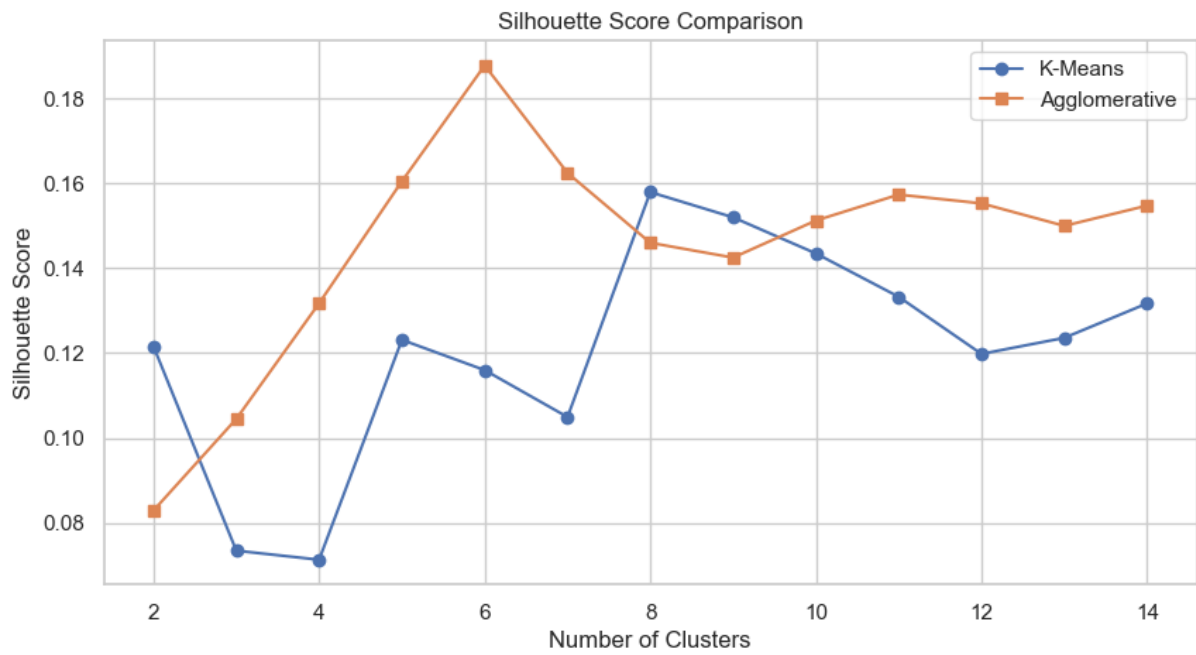


Figure 8.8: Silhouette Score Comparison between Kmeans and Hierarchical clustering

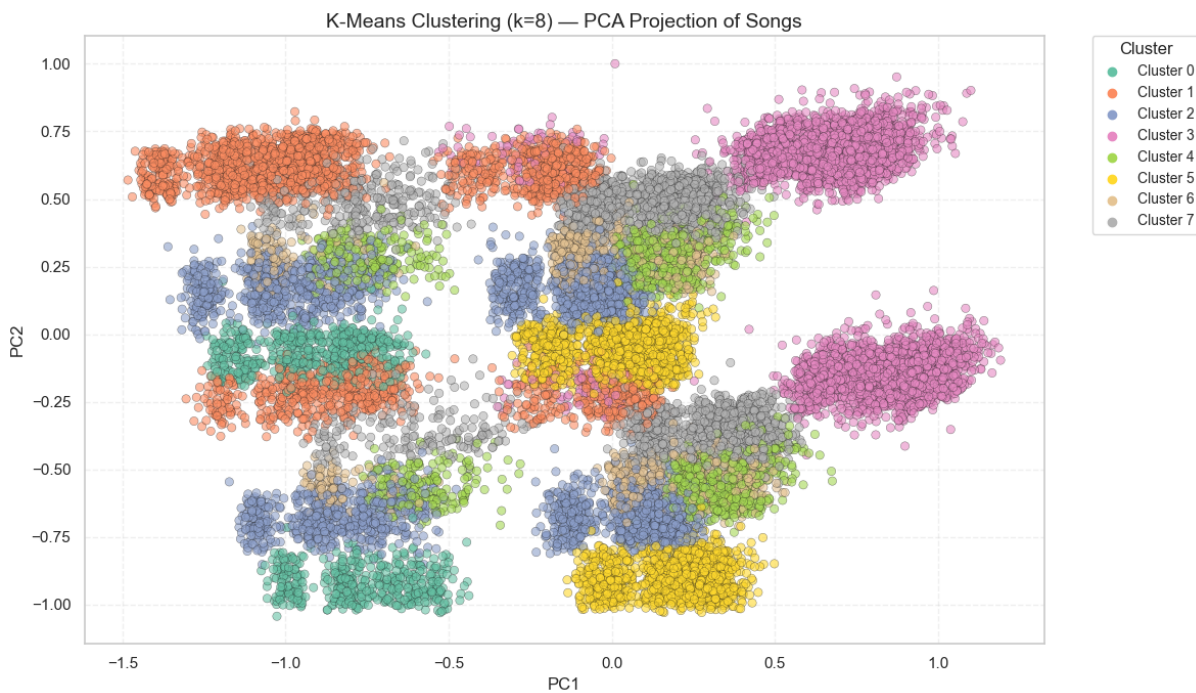


Figure 8.9: Kmeans Clustering



Figure 8.10: Kmeans Clustering (Using first 2 principal components/dimensions only)

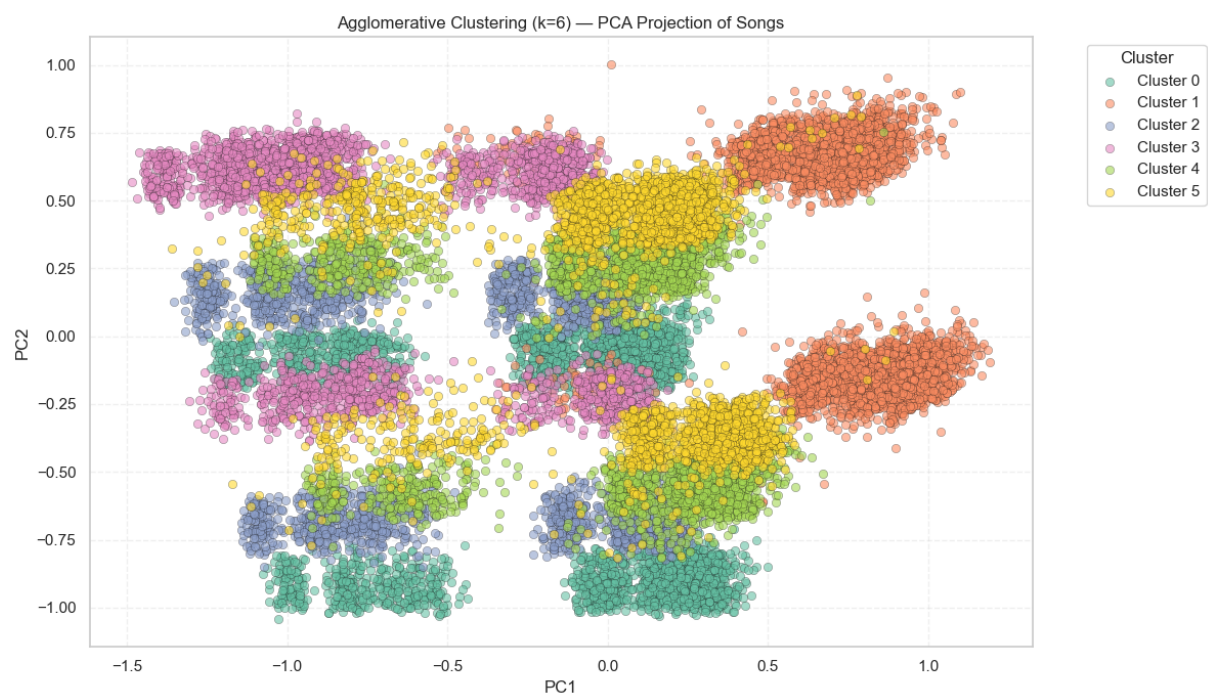


Figure 8.11: Hierarchical/Agglomerative clustering

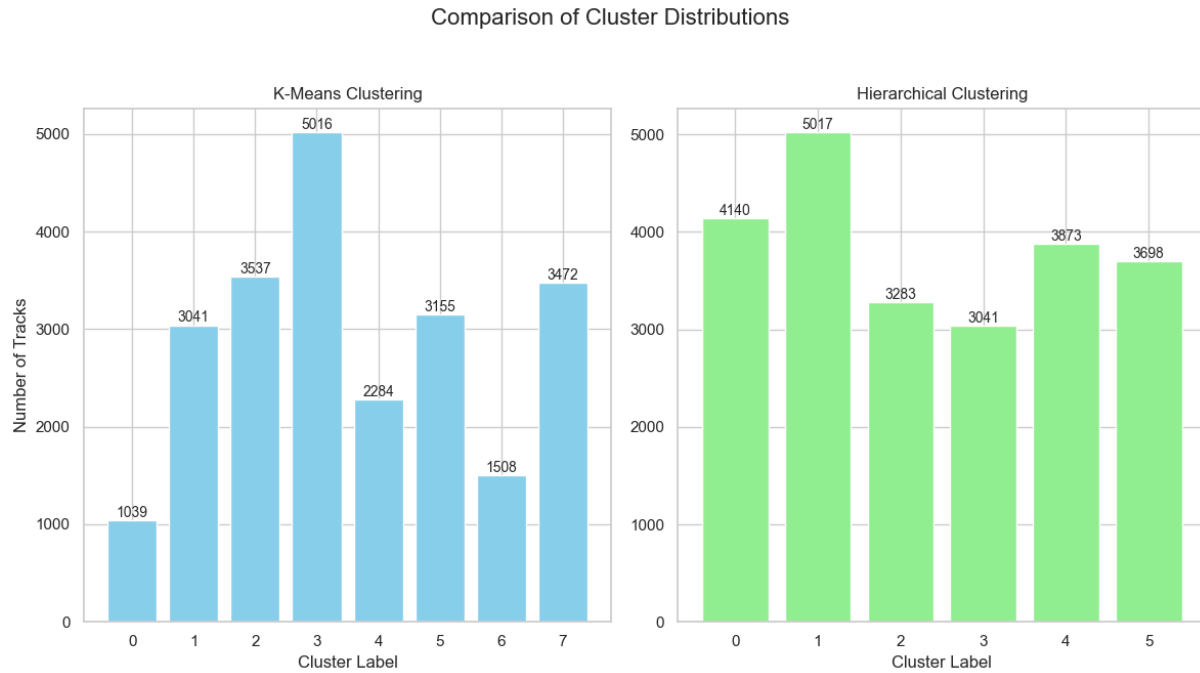


Figure 8.12: Clusters Distribution by Clustering Algorithm

7. Visualization

Throughout the project, visualization played a key role:

PCA Scatter Plots: Showed the distribution of tracks in low-dimensional space.

Scree Plots: Illustrated variance explained by each principal component.

Cluster Plots: Highlighted groupings identified by K-Means and hierarchical clustering.

Cluster Distribution Bar Plots: Compared the size of clusters between different algorithms.

6.2 Recommendation System (Hybrid KNN + CF) — Summary

Main Results:

- Hybrid recommender ($\alpha = 0.5$) gave best balance.
- KNN works well for new/niche tracks.
- Collaborative filtering excels for mainstream songs.
- PCA scatter plots reveal meaningful clusters.

Takeaway: Balanced discovery + familiarity $\rightarrow$ better engagement.

6.3 Clustering Insights — Summary

- K-Means produced 8 meaningful clusters.

- Hierarchical clustering found 6 broader mood/genre groups.
- Cluster distribution highlights underrepresented genres.

Chapter 9

Power BI Visualization

5.1 Overview

This chapter presents the visual analytics developed in Power BI to explore user churn behavior, song-level trends, and artist-specific performance metrics.

5.2 Users Analysis Dashboard

The churn dashboard focused on understanding the characteristics and behavior of users likely to cancel their subscriptions.

Key visuals included:

User Demographics and Distribution

- User count by subscription type and country → Identifies market distribution and dominant subscription tiers.
- Total number of users and countries → Provides dataset coverage overview.
- Users by gender and churn by gender → Highlights gender-specific churn trends.

Engagement Behavior

- Listening time (max, min, average) → Measures user engagement intensity.
- Average skip rate and listening time by age group → Examines age influence on patience and engagement.
- Users by device and subscription type → Shows platform usage preferences.
- Average ads per week and churn rate → Shows effect of advertisements on churn.

User Activity Insights

- Average songs played per day per age group.
- Ads per week vs. listening time.
- Total songs played per day by subscription type and churn status.

5.3 Songs Analysis Dashboard

The popularity dashboards for 2023 and 2025 visualize global musical and artist trends.

General Song and Artist Metrics

- Most popular artists (2025)
- Number of albums, tracks, artists, and genres
- Songs by genre and count of genres by season

Popularity and Audio Features

- Average popularity (2023 & 2025)
- Average popularity vs. instrumentalness
- Average popularity vs. valence
- Average popularity by release year

Genre and Period Analysis

- Count of tracks by genre and sub-genre
- Count of tracks by period (early 2000s, 2010s, 2020s)
- Average popularity per key, genre, loudness, energy

Song-Level Insights

- Song duration (max, min, avg)
- Avg popularity by song name
- Songs and albums per artist

5.4 Artist-Specific Dashboards

Eminem Dashboard

Main Visuals:

- Average, max, min popularity
- Number of songs and albums
- Popularity comparison by year
- Top 10 most popular songs
- Songs and albums by year

Key Findings:

- Peak popularity in 2006

- 12 albums and 316 songs
- Popularity spikes align with major releases
- Success tied to lyrical intensity and consistent output

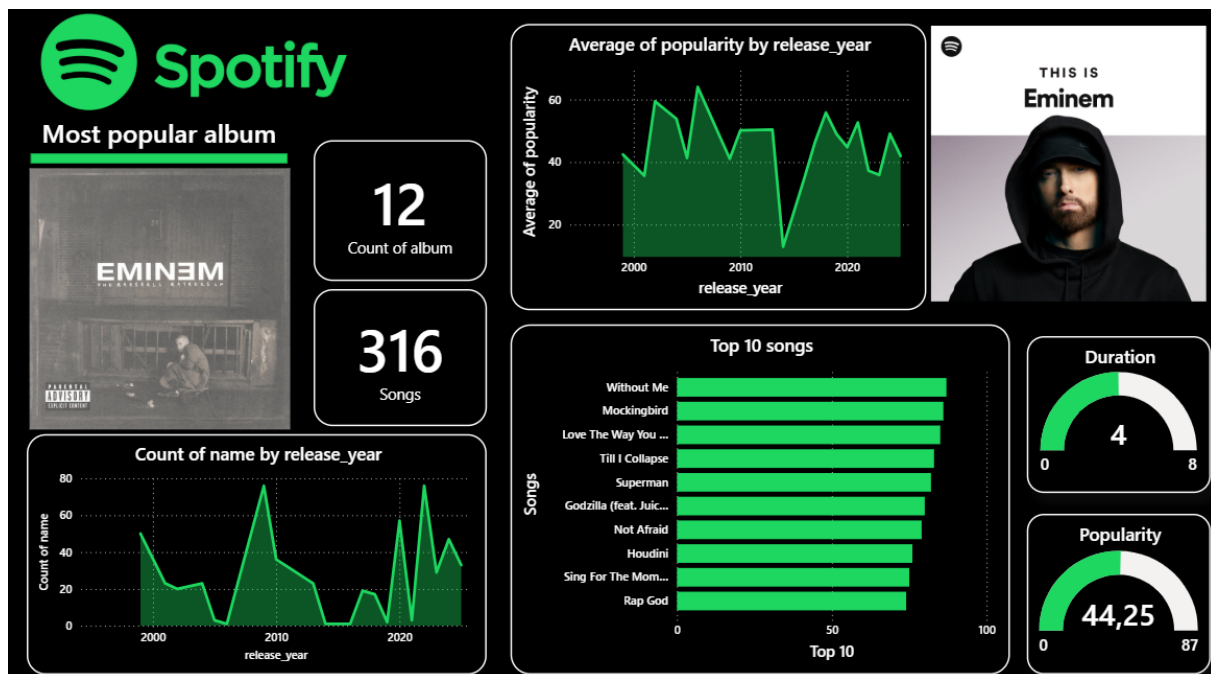


Figure 9.1: Eminem Comprehensive Dashboard

NF Dashboard

Main Visuals and Findings:

- Peak popularity in 2017 (Perception)
- 4 of top 10 songs come from Perception
- 6 albums and 54 songs
- Strong emotional themes maintain engagement
- Popularity renewed in 2023



Figure 9.2: NF Comprehensive Dashboard

Chapter 10

Insights & Takeaways

7.1 User Churn Insights

- High churn among Free users (ads + restrictions).
- Premium users show highest satisfaction.
- Younger users skip more and engage less.
- More ads → higher churn probability.

Actionable Recommendation: Target Free mobile users with personalized Premium offers.

7.2 Popularity Prediction Insights

- CatBoost performed best.
- Popularity strongly driven by:
 - Energy, valence, danceability
 - Artist reputation
 - Recency

7.3 Recommendation Insights

- Hybrid recommender most effective.
- Clusters align with moods and genres.

7.4 Artist Insights

- Eminem shows sustained long-term popularity.
- NF peaked in 2017 but rebounded in 2023.
- Arabic market dominated by few top artists.

Chapter 11

Conclusion & Future Work

This project integrates business intelligence, predictive modeling, and recommendation systems to deliver a comprehensive data-driven understanding of the Spotify ecosystem. The framework enhances user retention, improves song discovery, and enables deeper insights into global music consumption behavior.

As a future work, we can integrate the developed system into a web application that includes, attractive CSS styling, Interactive features for song selection and parameters tuning, automated ETL/ELT process using cloud solutions. We can also contrive an other new feature within the web application to display a dashboard that shows key metrics and statistics related to a specific artist, song, album, genre, etc... that is based on the same aforementioned ELT pipeline.